

POWER BI MODULE END PROJECT - DOCUMENTATION

Module-End

Power BI Module End Project – Logistics &
Transportation

Project Title

Bike Sharing Stations Data Analysis using Power BI

Submitted By

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ANB18 Batch

Project Overview :

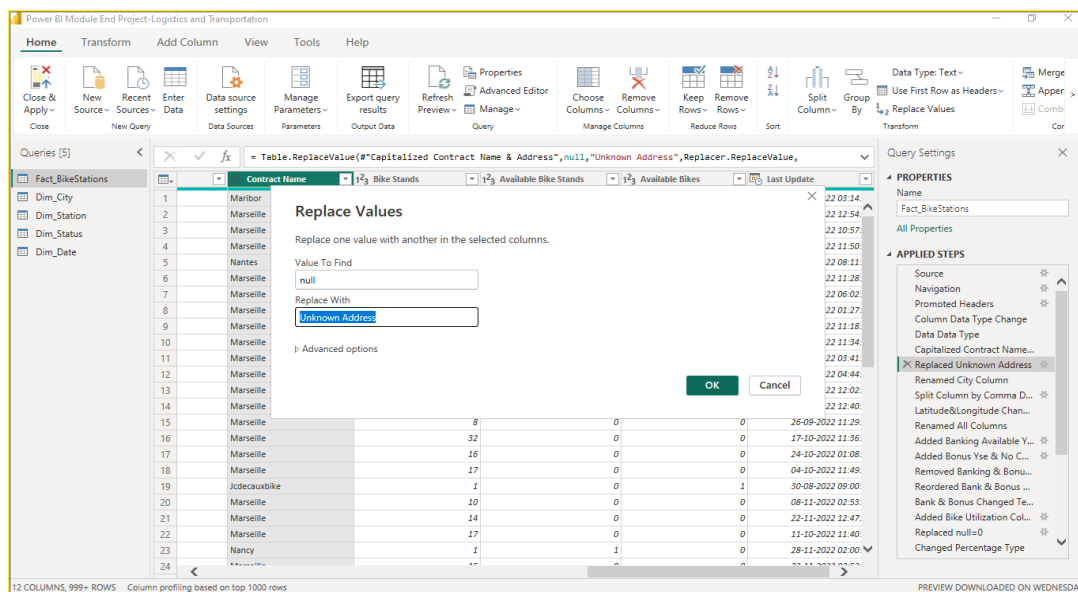
- This report documents the end-to-end Power BI solution developed as part of the Logistics and Transportation module project. The dashboard provides comprehensive visibility into bike-sharing station operations across multiple European and Asian cities, enabling data-driven decision-making for fleet management and station planning.

Objectives :

- Monitor total bike fleet availability and utilization across all cities.
- Track station open/closed status and stand occupancy in real time.
- Identify cities with the highest bike demand and utilization rates.
- Analyze year-over-year growth in available bike inventory
- Provide interactive filtering by city, station status, update year, and update month.

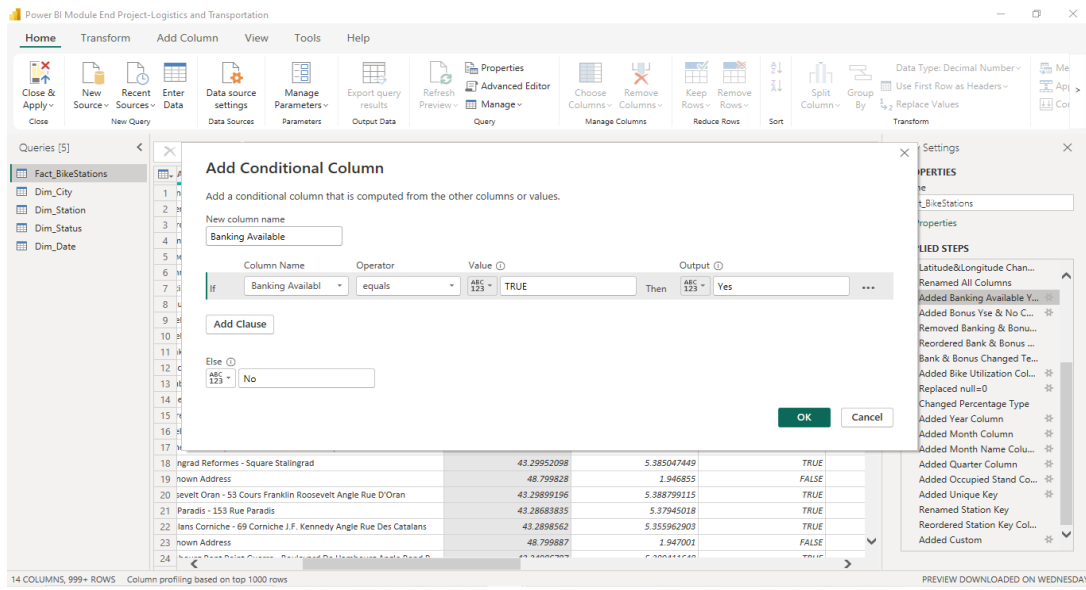
Power Query :

Replace Value



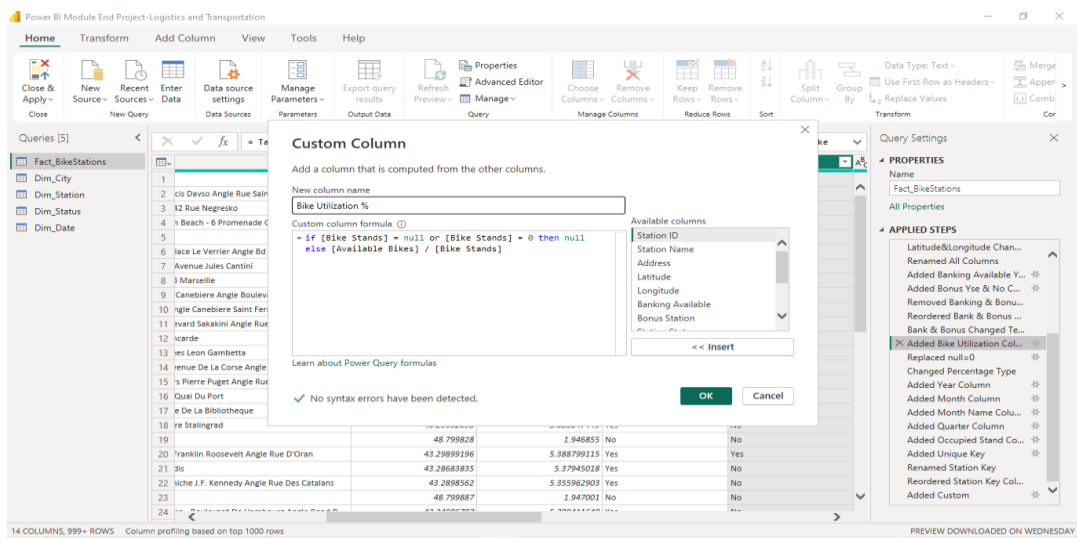
- ❖ I have fully analyzed and cleaned my dataset using Power Query. During the process, I performed several data transformation steps, including standardizing text casing and replacing null values in the address fields with 'Unknown Address'.

Conditional Column



- ❖ Subsequently, I extracted the relevant columns using the Delimited Column feature in Power Query. I also generated the required columns by applying conditional logic.

Custom Column

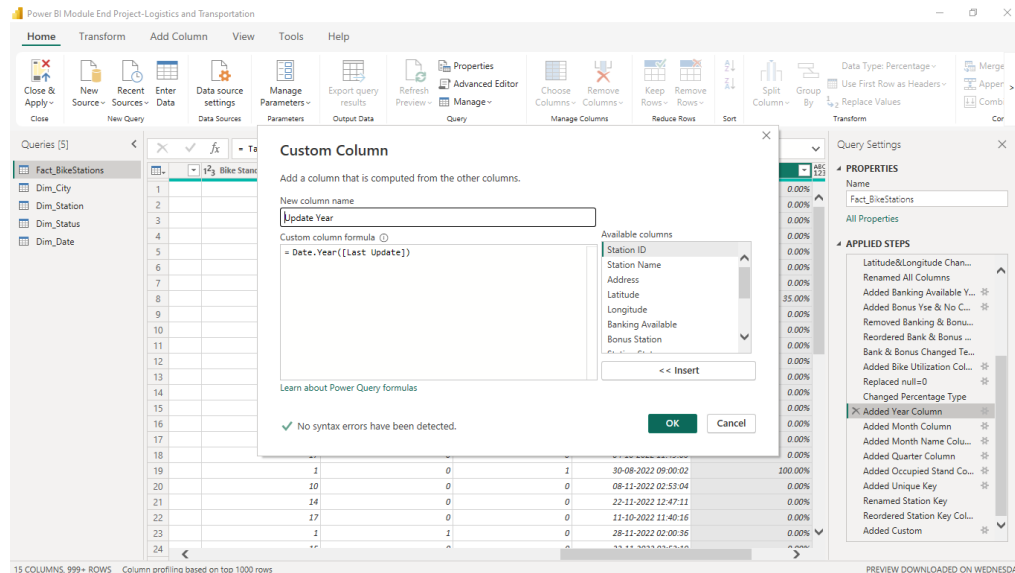


- ❖ Since bike utilization percentage was a key metric for my analysis, I created a Custom Column in Power Query using the formula below. I then converted the resulting column's data type to Percentage, enabling clear and accurate measurement of bike utilization across stations.

Formula : = if [Bike Stands] = null or [Bike Stands] = 0 then null else [Available Bikes] / [Bike Stands]

Formula : = [Bike Stands]-[Available Bike Stands]

Date, Month, Month Name, Quarter, Year



- ❖ Next, I needed a separate Date column for time-based analysis. Using the 'Last Update' field, I created a Date column and then extracted the Year, Month, Day, and Quarter components from it. The formula used for this transformation is provided below.

Formula : = Date.Day ([Last Update])

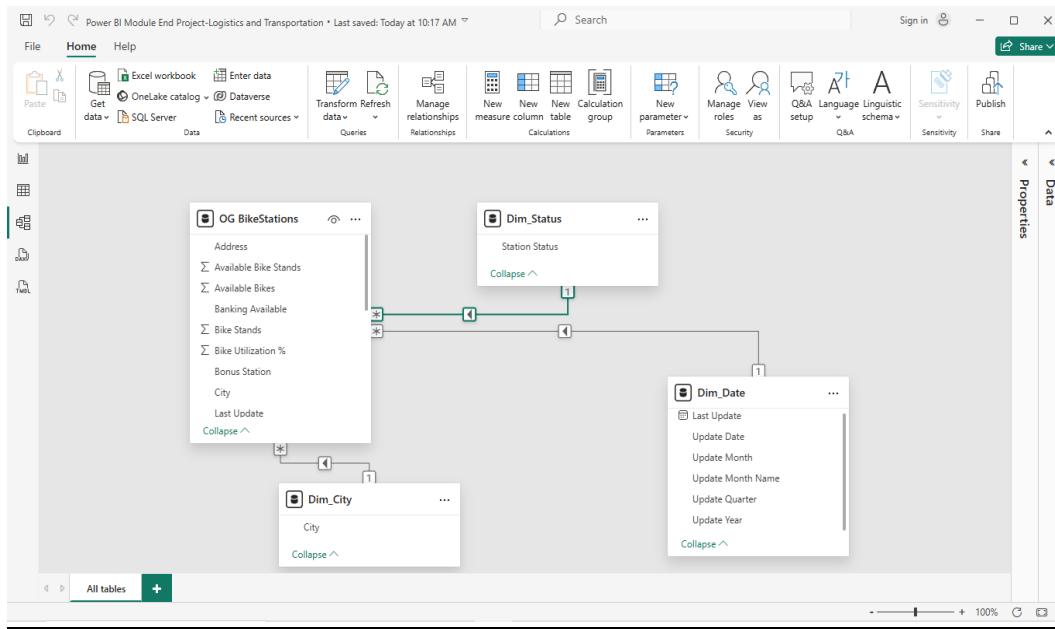
Formula : = Date.Month([Last Update])

Formula : = Date. MonthName ([Last Update])

Formula : = Date.QuarterOfYear ([Last Update])

Formula : = Date.Year([Last Update])

DATA MODELLING



- ❖ To build a star schema data model, I extracted relevant columns from the 'Fact_Bike_Station' table and created separate dimension tables: 'Dim_City', 'Dim_Status', and 'Dim_Date'. These tables contain unique records and were used to establish relationships with the fact table, improving data organization and analytical performance.

DAX

Station ID	Station Name	Address	Station Status	City	Bike Stands	Available
3071	BIOPARC	Angle Ave Rockefeller / Cours Eugénie Face Au Bioparc	OPEN	Lyon	20	
134	C/ MAXIMILIANO THOUS	Maximiliano Thous - Luz Casanova	OPEN	Valence	20	
118	C/ DR. GOMEZ FERRER	Gómez Ferrer - Álvaro De Bazán	OPEN	Valence	20	
201	AVDA. PEREZ GÁLDOS	Perez Galdós - Nou Moles	OPEN	Valence	20	
202	PASEO PECHINA	Pechina - Teruel	OPEN	Valence	20	
136	CALLE ECONOMISTA GAY	Economista Gay - Constitución	OPEN	Valence	20	
43	C/ DE ORIENTE	Escultor José Capuz - Oriente	OPEN	Valence	20	
86	AVDA. GASPAR AGUILAR, ESQ. VICENTE PARRA	Gaspar Aguilar - Vicente Parra	OPEN	Valence	20	
152	00152 - THÉÂTRE DES MAZADES	10 Av Des Mazades	OPEN	Toulouse	20	
101	00101 - LESPINET - MARNE	Place De Lespinet - Face 51 Avenue De Lespinet	OPEN	Toulouse	20	
6032	PRÉFECTURE / SARRAIL	Angle Rue Vauban	OPEN	Lyon	20	
178	CALLE REIG, GENOVES	Reig Genovés - Ramón Contreras Mongrell	OPEN	Valence	20	
74	POVSETOVA - KAJUHOVA	Povsetova - Kajuhova	OPEN	Ljubljana	20	
151	CALLE URQUIZA	Calle Urquiza - Aprox. C/ Venecia	OPEN	Seville	20	
74	PLAZA SAN FELIPE NERI (Mercado Algrós)	Plaza San Felipe Neri (Mercado Algrós)	OPEN	Valence	20	
30	AVENIDA EDUARDO DATO	Avenida Eduardo Dato - Detrás Estadio Sánchez Pizjuan	OPEN	Seville	20	
17	GOLDEN LANE	Golden Lane	OPEN	Dublin	20	
184	00184 - MICHOUN - HYÈRES	70 Chemin Michoun	OPEN	Toulouse	20	
141	C/ GREGORIO GSA (PEATONAL)	Gregorio Gsa - Padre Ferris	OPEN	Valence	20	
160	JOSE MARIA DE HARO	José María De Haro - Justo Y Pastor	OPEN	Valence	20	
53	R00053 - LUDWIG VAN BEETHOVEN	- Ludwig Van Beethoven - Rue Ludwig Van Beethoven (Devant N° 45)	OPEN	Luxembourg	20	
89	R00089 - PLACE THORN	Place Thorn/Bd Marcel Cahen - 50M Avant Rue Béatrix	OPEN	Luxembourg	20	
231	00231 - NARBONNE - DURAND	Face 215 Rte De Narbonne Angle All Emile Durand	OPEN	Toulouse	20	
100	00100 - PORT ST-SAUVEUR - SABATIER	16 All Paul Sabatier	OPEN	Toulouse	20	

- ❖ Once the data preparation phase was completed and the cleaned dataset was loaded into Power BI, I developed additional DAX measures to derive key performance indicators (KPIs). These measures enabled deeper insights into bike station performance and supported the dashboard's analytical requirements. The corresponding DAX formulas are provided below.

DAX FORMULA

Formula : Available Stands = SUM('OG BikeStations' [Available Bike Stands])

Formula : Number of Stations = DISTINCT COUNT('OG BikeStations' [Station Name])

Formula : Occupied_Stands = SUM('OG BikeStations' [Occupied Stands])

Formula : Open Stations = CALCULATE (COUNTROWS('OG BikeStations'), 'OG BikeStations' [Station Status] = "OPEN")

Formula : Total Bikes = SUM('OG BikeStations' [Available Bikes])

Formula : Total Stands = SUM('OG BikeStations' [Bike Stands])

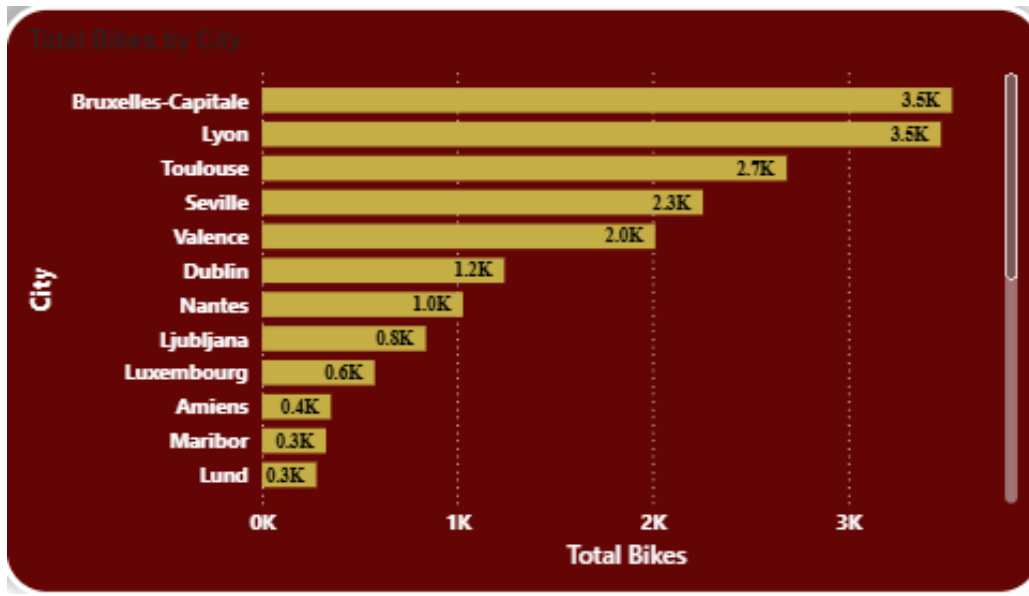
Formula : Utilization Rate = DIVIDE (SUM('OG BikeStations' [Available Bikes]), SUM('OG BikeStations' [Bike Stands]))

POWER BI VISUALISATIONS

- ❖ After completing the data cleaning and DAX calculations, I used KPI Charts to create visualizations based on the measures. Here are the key charts I have generated.

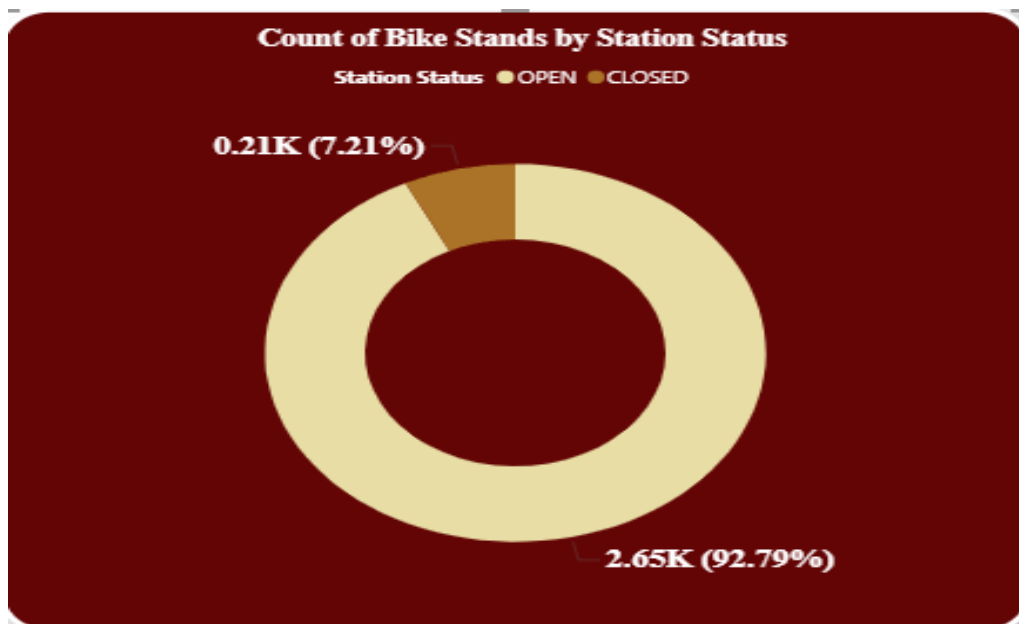


Clustered Column Chart



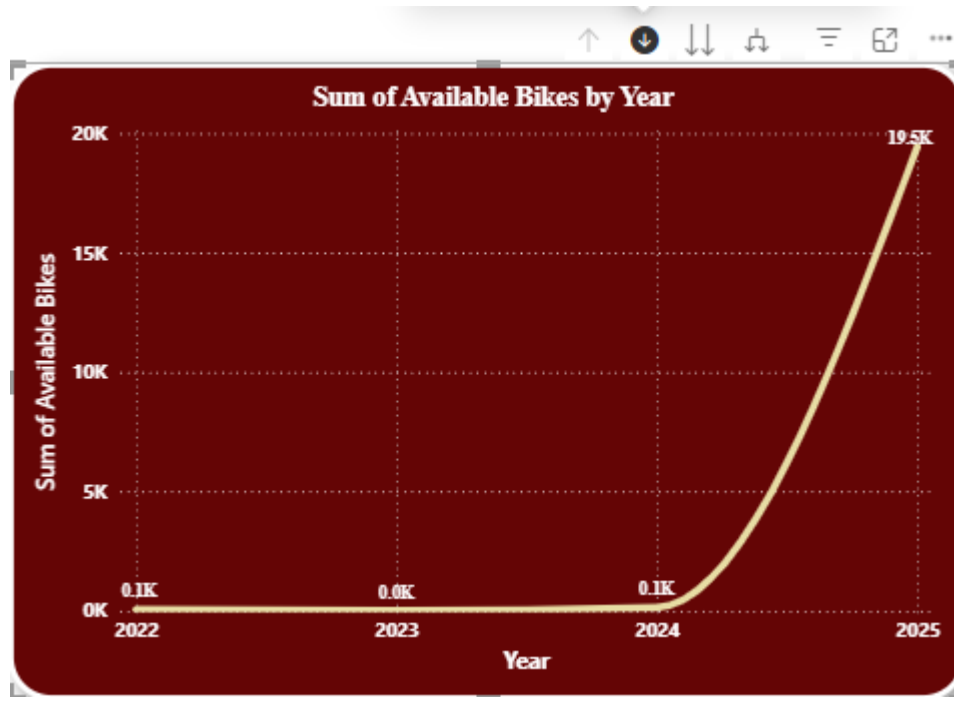
- ❖ When analyzing the total number of bikes by city using a clustered column chart, Bruxelles-Capitale ranks first

Donut Chart



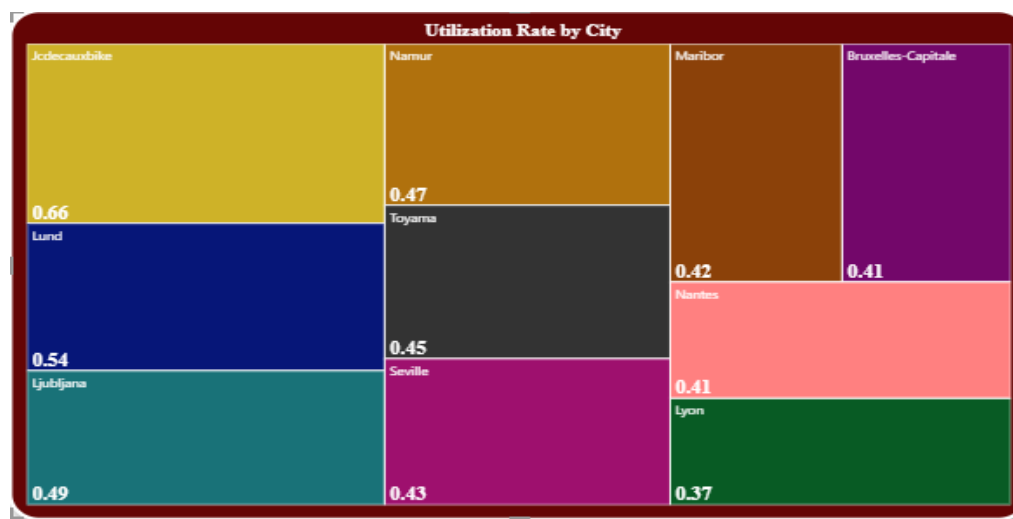
- ❖ Next, I used a donut chart to determine whether the bike stands were open or closed; the chart shows that approximately 92.79% of the bike stands are open.

Line Chart



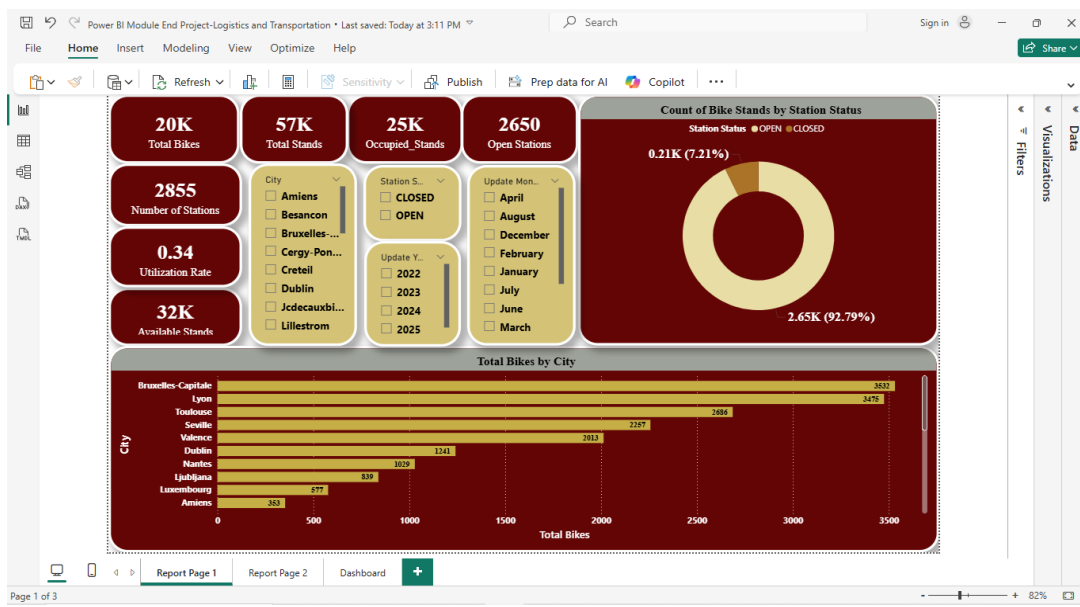
- ❖ I used a line chart to track the total number of bicycles available annually. This data covers the period from 2022 to 2025; notably, a comparison of the early part of 2025 with the years 2022, 2023, and 2024 reveals a higher number of bicycles available for use. Furthermore, since the 'Date Hierarchy' feature has been utilized, the 'Drill-through' option allows us to view the data by year, month, or quarter.

Tree Map

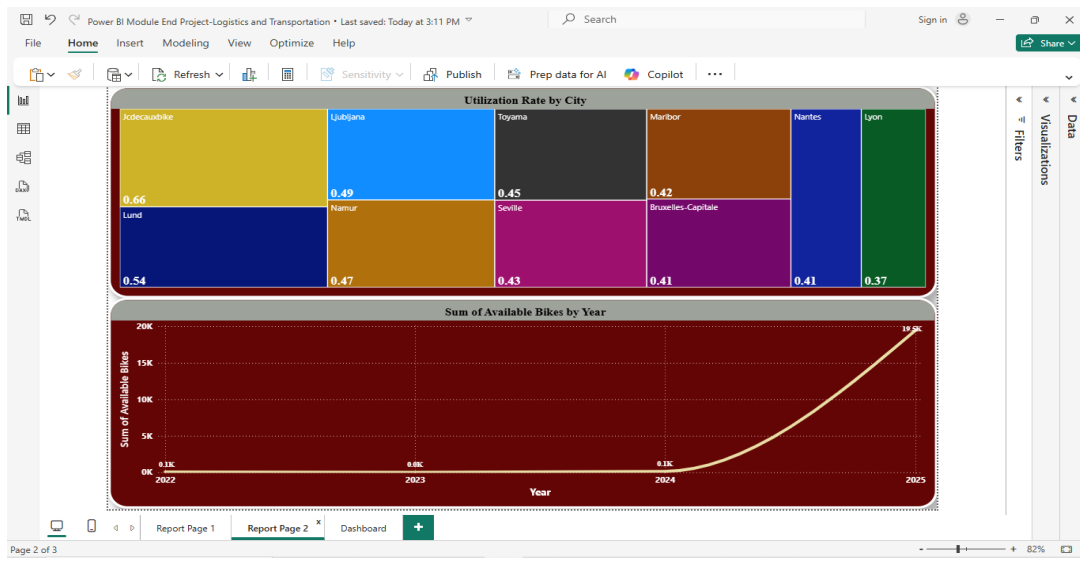


- ❖ Finally, I used a treemap to identify the top 10 cities with the highest bike-sharing station utilization. To achieve this, I applied a filter to select the 'Top 10' option for the city-wise analysis and visualized the results on the Treemap.
- ❖ I have added the necessary slicers - such as City, Station, Status, Year, and Month - which make the visualizations much easier to understand, allowing us to view and interpret the data in various ways: by city, year, month, or status.

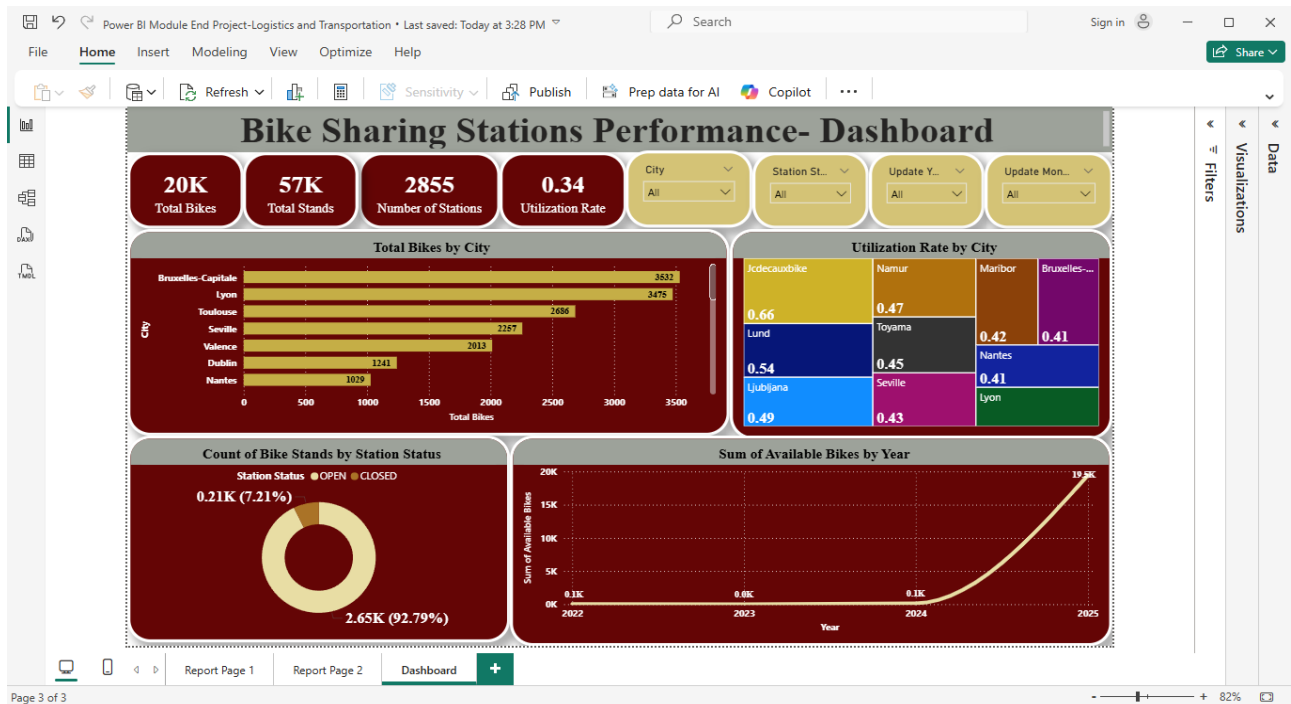
Visuals Report Page-1



Visuals Report Page-2



DASHBOARD



- ✓ A Scalable, Data-Driven Monitoring Solution Was Successfully Built—covers the dashboard scope, visuals used, and stakeholder accessibility
- ✓ Network Performance Is Strong but Unevenly Distributed — highlights the 92.79% open rate vs. the utilization gap between cities like Jcdecauxbike (0.66) and Lyon (0.37)
- ✓ The 2025 Growth Spike Signals a Critical Expansion Phase — addresses the near-zero to 19.5K available bikes jump and what it implies operationally
- ✓ The Dashboard Provides a Strong Foundation for Continuous Improvement — speaks to the slicer framework's adaptability and future integration potential

Thank You!